

ouees-202606 topic 01:

- **Internetworking behind AI**
- **or how to build the gigantic computing infrastructure**

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On the internet

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Lecture notes and reporting

- <https://github.com/jj1bdx/oueees-202606-public/>
- Check out the README.md file and the issues!
- Keyword at the end of the talk
- URL for submitting the report at the end of the talk

Topic of this video:

Internetworking behind AI

What we call AI is a computing service

- ChatGPT, Claude, Gemini are NOT on a single computer; they are *distributed*
- AI is a collective service running on *massive number of clusters of computers*
- Localized AI exists, but the scale is smaller than commercial AI services

What do AI services actually?

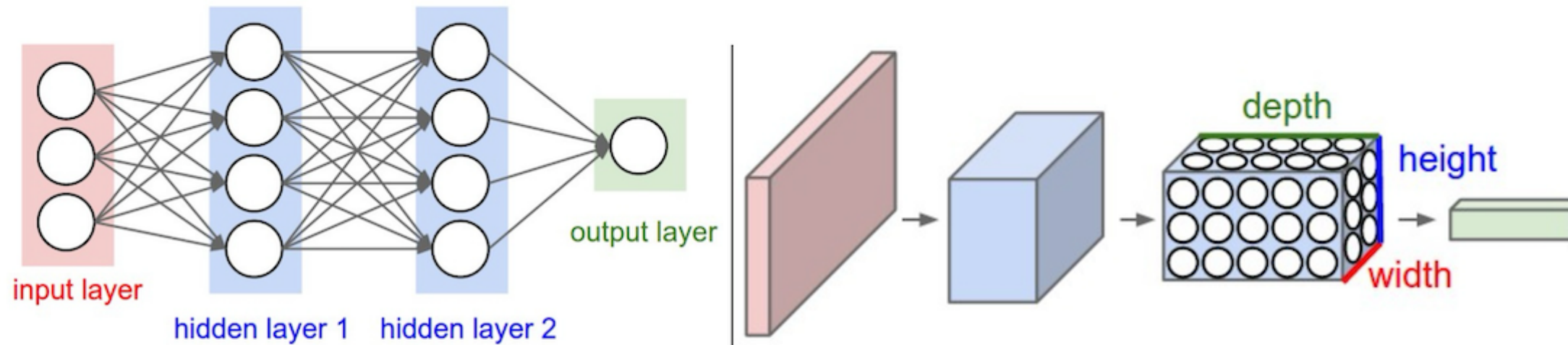
- They compute huge models of neural networks of an undisclosed scale
- They run Large Language Models (LLMs)
- Open-source models include 1-trillion parameters with active 30- to 50-billion ones ¹

¹ Carolyn Weitz, "Best Open Source LLMs: Benchmarks, Licenses, Local Deployment and Enterprise Use Cases", AceCloud, Last Updated: May 13, 2026

What does AI compute for the neural networks?

- Linear algebra + statistical functions
- Equivalent to huge ($> 1\text{B} \times 1\text{B}$) matrix computation
- Randomization during the computation involves
- *Non-deterministic output for the same input*

Convolution by multiple neural networks ²



Left: A regular 3-layer Neural Network. Right: A ConvNet arranges its neurons in three dimensions (width, height, depth), as visualized in one of the layers. Every layer of a ConvNet transforms the 3D input volume to a 3D output volume of neuron activations. In this example, the red input layer holds the image, so its width and height would be the dimensions of the image, and the depth would be 3 (Red, Green, Blue channels).

² **Convolutional Neural Networks (CNNs / ConvNets), CS231n: Deep Learning for Computer Vision, Stanford University, Spring 2026**

Large Language Models (LLMs)

- A neural network trained on a vast amount of text for natural language processing tasks, *especially language generation* ³
- Generative pre-trained transformers (GPTs): no recurrent pre-trained on large datasets of unlabeled content, and able to generate novel content ⁴

³ **Large language model**, Wikipedia, (last visited June 16, 2026), emphasis by Kenji Rikitake.

⁴ **Generative pre-trained transformer**(last visited June 16, 2026).

So how do you run an LLM on computers?

- GPUs with large-scale memories and high bandwidths
- Example: NVIDIA H100 Tensor Core GPU⁵
 - Architecture: Hopper
 - Memory: 80GB HBM3
 - Memory Bandwidth: 3.35 TB/s
 - Tensor Performance: Up to 3,958 TFLOPS (FP8)
 - Power: 350-400W (NVL) / 700W (SXM)

⁵ Deborah Emeni, 12 best GPUs for AI and machine learning in 2026, Northflank

Massive computing needs massive resources

- Each LLM inference occupies an entire GPU
- Electric power to run the GPU servers
- Water and electricity to *dissipate heat*
- Land to place the servers for the *data centers*
- Internet bandwidth to utilize the GPUs

High bandwidth needs shorter latency and mass interconnects between distributed servers

- Minimize latency locally, regionally and globally
 - Data center building rush in many countries
- Interconnecting servers *within* the data center
 - Optic fiber networks (~400Gbps)
 - Highly-efficient protocols *other than* Ethernet and TCP/IP
 - Backplane interconnect: up to 112Gbps and 50GHz ⁶

⁶ High-Speed Backplane Interconnect Solutions Quick Reference Guide, TE Connectivity (PDF)

AI needs fast database and network servers

- AI needs fast and large-scale data and database**
- AI needs fast and large-scale Web servers**
- AI needs massive and reliable internet connection**
- AI raises the demand of high-performance computing as a whole**

Running AI services is a security issue

- AI services are part of critical national infrastructure
- Military systems partly run on commercial AI
- Cyberattacks can paralyze emergency services
 - Ransomware effectively kills hospital operation
 - US government halted Claude Mythos and Fable 5 model services for "*non-nationals*"

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